

Chromic Vertisols:

Chromic Vertisols are clay-rich soils ($\geq 30\%$ clay, often swelling clays like smectites) that display strong shrink-swell behavior. The "chromic" qualifier indicates a distinct reddish or brownish color due to iron oxides. They form deep, wide cracks during dry seasons, and slickensides or wedge-shaped aggregates are common.

Drainage:

- Generally **imperfectly to poorly drained** because of slow permeability in wet conditions.
- During dry seasons, cracks improve infiltration but also cause uneven wetting.
- Temporary waterlogging can occur in flat landscapes.

Depth:

- Usually very deep (>100 cm).
- Rooting is often restricted by shrink-swell movements (shearing roots) and alternating wet/dry conditions.

Workability:

- Vertisols are among the most difficult soils to manage mechanically because of their strong shrink-swell properties.
- When wet: they are sticky, plastic, and heavy — machinery easily gets stuck, and tillage smears and compacts the soil.
- When dry: they harden into very dense clods or crusts that are almost rock-like, making ploughing and seedbed preparation very difficult without heavy equipment.

Nutrient Richness & Cation Exchange Capacity (CEC):

- Chromic Vertisols have a High CEC (30 and 80 cmol(+)/kg of dry soil) and High Base saturation(50-100%).
- Has a high Ca^{2+} and Mg^{2+} content. The Ca/Mg- ratio is normally between 3 and 1.
- Deficient in Nitrogen and organic Matter.
- Low availability of Phosphorus.
- Contain much exchangeable aluminium and are low on sulphur and/or zinc.

pH:

- Usually neutral to alkaline (pH 6.5–8.5).
- Rarely acidic due to high base content.

Management:

- Promote contour ploughing, bunding, tied ridges, and vertical mulching to enhance infiltration and conserve water, particularly on slopes.
- Use deep tillage (chiseling or subsoiling) to break hard horizons and improve rooting.
- Maintain vegetation cover or grass strips on bunds and risers to reduce runoff velocity and soil loss.
- Incorporate organic matter (compost, manure, crop residues) to improve soil structure and biological activity.
- Use balanced fertilization, especially nitrogen and phosphorus, applied in split doses to minimize losses.
- Carry out tillage only within the narrow workable moisture window (slightly moist, not too wet or dry).
- Favor crops tolerant to variable moisture (cotton, wheat, barley, sorghum, chickpeas, flax and sugar cane). **Cotton** performs well on Vertisols because cotton has a vertical root system that is not severely damaged by cracking of the soil.